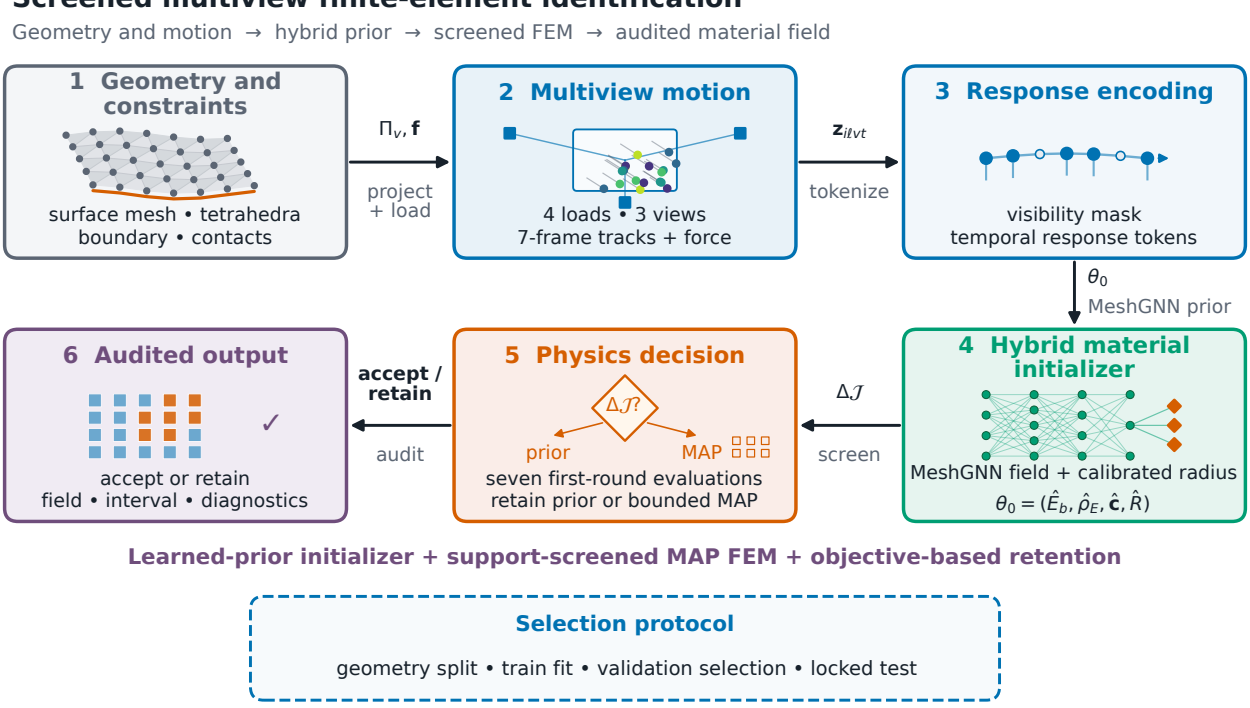
Screened multiview finite-element identification
Geometry and motion → hybrid prior → screened FEM → audited material field
1 Geometry and constraints
surface mesh • tetrahedra
boundary • contacts
$\Pi_v, \mathbf{f}$
project + load
2 Multiview motion
4 loads • 3 views
7-frame tracks + force
$\mathbf{z}_{i\ell vt}$
tokenize
3 Response encoding
visibility mask
temporal response tokens
θ_0
MeshGNN prior
4 Hybrid material initializer
MeshGNN field + calibrated radius
$\theta_0 = (\hat{E}_b, \hat{\rho}_E, \hat{\mathbf{c}}, \hat{R})$
$\Delta\mathcal{J}$
screen
5 Physics decision
$\Delta\mathcal{J}$?
prior
MAP
seven first-round evaluations
retain prior or bounded MAP
accept / retain
audit
6 Audited output
accept or retain
field • interval • diagnostics
Learned-prior initializer + support-screened MAP FEM + objective-based retention
Selection protocol
geometry split • train fit • validation selection • locked test